

Task 1.

Using commands “meshgrid”, “surf”, “xgrid”, plot surface

$$z(x,y)=(1+2x)\sin(2x+1.5y-2)+0.5y(y-2/3)$$

in the rectangle $-5 \leq x \leq 2$, $-2 \leq y \leq 2$.

Press and retain the right button of mouse to rotate the image and see where the minimum of z is.

Task 2.

Ignoring Scilab and Excel, and using only a pen and piece of paper, find equation of a straight line that provides the least-squares approximation of the function given in table:

$x_0 = 1$	$x_1 = 2$	$x_2 = 3$
$y_0 = 3$	$y_1 = 5$	$y_2 = 6$